



Shengxian decoction alleviates experimental autoimmune myasthenia gravis by enhancing the immunosuppressive activity of regulatory T cells via Hippo pathway

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ABSTRACT

Ethnopharmacological relevance: Myasthenia gravis (MG) is an autoimmune disease mediated by autoantibodies, the pathogenesis of which is well defined but for which safe and effective drugs are still lacking. Shengxian Decoction (SXD) is a classical Chinese herbal formula that has been shown to improve qi deficiency and fatigue syndrome, which is also one of the key pathomechanisms in myasthenia gravis. However, the potential of this formula to treat myasthenia gravis (MG) and its underlying mechanisms remains to be fully elucidated.

Aim of the study: To investigate the therapeutic effect of the traditional Chinese medicine formula, Shengxian Decoction (SXD) on MG and its potential mechanism of action.

Methods and materials: Employing prednisone (PDN) as a positive control, the experimental autoimmune myasthenia gravis (EAMG) rat model induced by recombinant acetylcholine receptor (AChR) α subunit 97–116 peptide was used to reveal and confirm the therapeutic effect of SXD on MG. Model evaluation was performed based on animal behavioral performance combined with repeated nerve stimulation (RNS) experiments. Hematoxylin-Eosin (HE) staining and enzyme-linked immunosorbent (ELISA) assay were used to detect the level of inflammatory cytokines. The alterations in T-cell immunity were observed using flow cytometry (FCM). Immunohistochemistry (IHC) and immunofluorescence (IF) experiments were used to detect the molecular expression of TAZ, TEAD, and FOXP3. The ex vivo and in vivo mechanisms were validated by Western blot (WB) and Co-immunoprecipitation (Co-IP).

Results: The results showed that SXD significantly reversed loss of weight and muscle strength in EAMG rats, reducing the inflammation level and repairing the damage of thymic tissues in the model rats. Furthermore, SXD altered the trend observed in EAMG rats by markedly increasing the proportion of pro-inflammatory Th1 and Th17 cells and reducing the expression of Regulatory T cells (Treg), which plays a pivotal role in regulating the immune response in EAMG rats. SXD was shown to upregulate the expression of FOXP3 and TEAD, downregulate the expression of TAZ, and inhibit the binding of TAZ and TEAD in this research. This phenomenon was subsequently validated in vitro, which may elucidate the potential mechanism of SXD in the treatment of EAMG.

Conclusion: SXD has been demonstrated to possess the capacity to treat EAMG. The underlying mechanism of action may be achieved by promoting the degradation of TAZ through the TAZ/TEAD pathway and inhibiting the binding of TAZ and TEAD, thereby restoring the balance of the body's immune function.

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1. Introduction

Myasthenia gravis (MG) is an autoimmune disease with cellular immune-dependent impairment of signaling at the neuromuscular junction mediated by antibodies to the acetylcholine receptor, which is prevalent in the elderly (Gilhus, 2016). The main feature of MG is generalized skeletal muscle fatigue intolerance, with respiratory and swallowing difficulties in severe cases (Gilhus et al., 2019). Although MG is considered to be an autoimmune disease with relatively well-defined antigenic antibodies, the current mainstream drugs, represented by prednisone, are still unable to meet the therapeutic needs and cannot prevent the side effects of the drugs that occur after long-term administration (Truhan and Ahmed, 1989). Therefore, it is important to search for safe and effective drugs for the treatment of MG.

Immune dysregulation is common in MG patients (Dasgupta et al., 1991; Wang et al., 2024). The thymus is an important immune organ in which T cells differentiate and develop. Studies have shown that abnormalities can be clearly seen in the thymic tissue of MG patients (Goldstein, 1971; Raica et al., 2008), with a long-term pathological state of chronic inflammation (Cron et al., 2018). Early-onset myasthenia gravis is often characterized by thymic hyperplasia, whereas late-onset myasthenia gravis is characterized by thymic atrophy, and in patients with benign thymomas, which are also associated with the manifestation of clinical symptoms of MG, thymic status and expression of genes associated with MG may directly reflect alterations in the pathogenic pathway (Fujii, 2013; Gilhus, 2016; Lee et al., 2020). It has been reported that interleukin 17 (IL-17), secreted by the Type 17 helper T cells (Th17), is able to act on self-antigens and trigger an autoimmune response in the body (Agalioti et al., 2023; Yasuda et al., 2019). The dynamic balance between pro-inflammatory Th17 cells and immunomodulatory Regulatory T cells (Treg) has become a focus of autoimmune disease research in recent years (Chen et al., 2021; Wang et al., 2023). A recent study found that the expression of the transcriptional co-activator TAZ is significantly increased during the differentiation and proliferation of Th17 cells (Geng et al., 2017). TAZ activity is regulated by the Hippo signalling pathway, a highly conserved signalling pathway that has been shown to be associated with cell proliferation and differentiation (Kwon et al., 2022; Li and Guan, 2022). When the Hippo pathway is activated, TAZ is degraded by phosphorylation in the cytoplasm (Ma et al., 2019). When the Hippo pathway is inhibited, TAZ enters the nucleus to activate and bind to the co-transcription factor TEAD, switching on the transcription of downstream genes, leading to abnormal cell proliferation (Hong et al., 2018).

Traditional Chinese medicine (TCM) has unique advantages in MG treatment due to its holistic immune regulation and fewer side effects. One study reported effective improvement of clinical symptoms in patients with myasthenia gravis after intervention with TCM alone without immunosuppressants (Qiu et al., 2024). It has also been reported that TCM can be effective in the regulation of mitochondrial energy metabolism and the repair of myogenic fiber damage in MG rats (Song et al., 2018). Shengxian Decoction (SXD) was first recorded in an ancient Chinese medicine book named *Yixuezhongzhongcaxilu* written by Zhang Xichun, a famous Traditional Chinese medicine practitioner in Qing dynasty. SXD consists of Astragali Radix (Huangqi, dried tubers of *Astragalus membranaceus* Fisch. ex Bunge), *Anemarrhenae Rhizoma* (Zhimu, dried tubers of *Anemarrhena asphodeloides* Bunge), *Bupleuri Radix* (Chaihu, dried tubers of *Bupleurum chinense* DC.), *Platycodonis Radix* (Jiegeng, dried tubers of *Platycodon grandiflorus* A.DC.), and *Cimicifugae Rhizoma* (Shengma, dried tubers of *Cimicifuga heracleifolia* Kom.) (All medicinal details refer to the Chinese Pharmacopoeia, botanical names verified by World Flora Online). SXD was recorded to treat a series of qi sinking symptoms such as lassitude, dyspnea, uterus and visceroptosis in *Yixuezhongzhongcaxilu*. Qi sinking syndrome is one of the main pathogenic mechanisms of MG, which results from the failure of the lifting or holding function of qi according to the basic theories of traditional Chinese medicine (Chen, 2023; Zuo et al., 2020).

Table 1
Chromatographic gradients.

Time (min)	A (%)	B (%)
1	98	2
5	80	20
10	50	50
15	20	80
20	5	95
27	5	95
28	98	2
30	98	2

SXD has the efficacy of ascending qi and elevating drooping, which has been commonly used in the treatment of MG for hundreds of years (Cao et al., 2024; Mao, 2017; Wang et al., 2021; Yao, 2019). The occurrence of inflammation and autoimmunity were recognized as pivotal process that contributes to the development of MG. Previous studies have shown that SXD can elevate TGF- β content, reduce IFN- γ and IL-17 content, lower serum AchR-Ab content, and reduce AchR damage at neuromuscular junctions, thereby ameliorating the clinical symptoms of myasthenia gravis (Wu et al., 2017; Xu et al., 2016; Zhu, J. et al., 2017). Nevertheless, its mechanism of action remains to be elucidated.

In this study, we successfully established an experimental autoimmune myasthenia gravis (EAMG) rat model using recombinant acetylcholine receptor (AChR) α subunit 97–116 (R97-116) peptide to induce autoimmunity in Lewis rats. The potential of SXD for the treatment of MG was demonstrated by behavioral manifestations combined with the repeated nerve stimulation (RNS) test. In addition, we observed that SXD was able to reverse the level of inflammation and the expression of proteins in the TAZ/TEAD pathway, thereby restoring the immune homeostasis of the organism. This study deepens the understanding of SXD for the treatment of myasthenia gravis, reveals the potential mechanism of therapeutic effects, and provides a new basis for the clinical application of SXD and future research.

2. Materials and methods

2.1. Drug preparation

All the traditional Chinese medicines of SXD were provided by the National Medical Hall of Nanjing University of Chinese Medicine. We weighed 180 g of *Astragali Radix*, 90 g of *Anemarrhenae Rhizoma*, 45 g of *Bupleuri Radix*, 45 g of *Platycodonis Radix* and 30 g of *Cimicifugae Rhizoma*, adding 1500 mL of water to completely submerge the herbs and soaking for 30 min, then decocted at a constant temperature of 100 °C for 40 min, filtered the impurities after cooling, then adding 1000 mL of water to the residue of the decoction and decocting for another 30 min. After filtration, the two decoctions were combined and concentrated to a final concentration of 1 g/mL using a rotary evaporator at constant temperature (Li et al., 2022).

Prednisone (Xinyi, Shanghai, China) was dissolved to a concentration of 0.54 mg/mL in distilled water.

2.2. Liquid chromatography-mass spectrometry (LC-MS)

The aqueous solution of the SXD sample was accurately quantified to 100 μ L. Subsequently, 300 μ L of a methanol solution was added, vortexed, and mixed for 10 min. The solution was then centrifuged at 13,000 rpm for 10 min, after which the supernatant was extracted for use in the subsequent experiments, which were conducted using an UltiMate 3000 RS chromatograph and a Q Exactive high-resolution mass spectrometer. The AQ-C18 column (150 $\times$ 2.1 mm, 1.8 μ m) was employed for the determination of the sample components, with a 0.1 % formic acid/water solution serving as the aqueous phase and methanol as the organic phase. Gradient elution is conducted in accordance with the specifications outlined in Table 1. The data acquisition time was 30

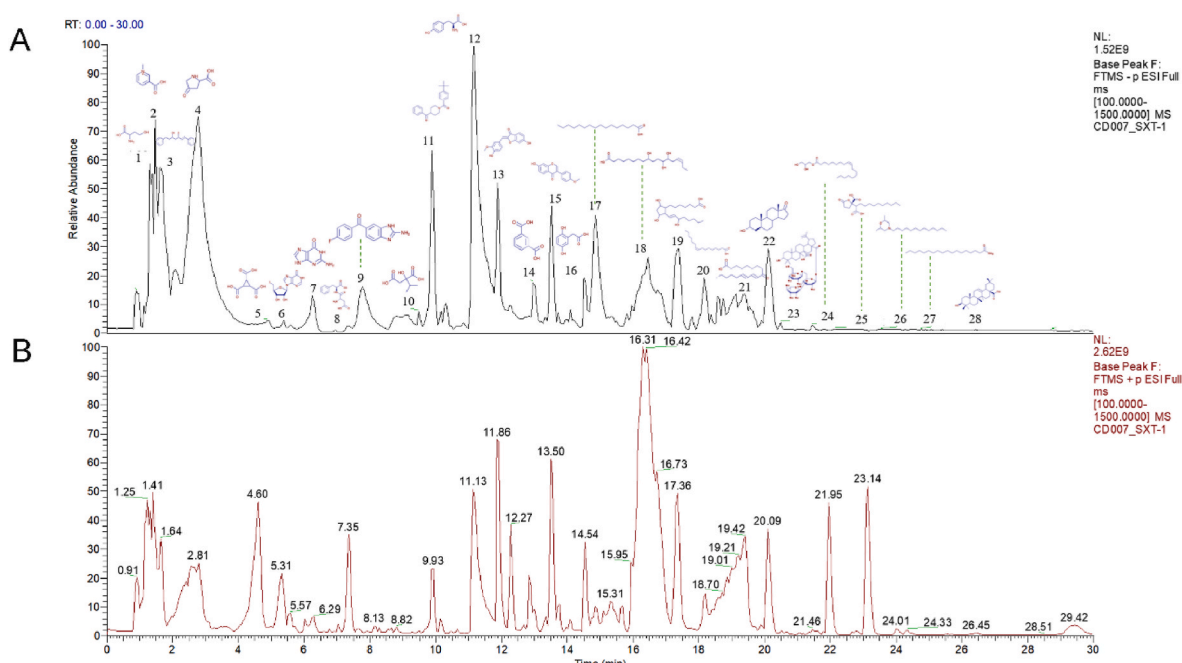


Fig. 1. LC-MS fingerprint chromatogram of SXD extracts solution.

A. Total ion flow diagram for negative ion mode.

B. Total ion flow diagram for positive ion mode.

Table 2

Results of LC-MS compositional identification of SXD.

NO.	RT (min)	Identification	Formula	Annotation MW	Calc. MW	Annot. DeltaMass	m/z
1	1.30	DL-Homoserine	C4H9NO3	119.05824	119.05686	-0.00138	118.04958
2	1.47	Trigonelline	C7 H7 N O2	137.04768	137.04762	-0.00006	138.0549
3	1.61	(1E)-5-hydroxy-1,7-diphenylhept-1-en-3-one	C19 H20 O2	280.14633	270.0963	-10.05003	303.12979
4	2.79	D-(+)-Pyroglutamic Acid	C5 H7 N O3	129.04259	129.04283	0.00023	130.05011
5	4.88	1,2,3-cyclopropanetricarboxylic acid	C6 H6 O6	174.01644	174.01546	-0.00098	173.00818
6	5.39	Inosine	C10 H12 N4 O5	268.08077	268.0808	0.00003	269.08798
7	6.26	Guanine	C5 H5 N5 O	151.04941	151.04841	-0.001	150.04114
8	7.32	α -Aspartylphenylalanine	C13 H16 N2 O5	280.10592	280.10607	0.00014	280.10607
9	7.78	2-Aminoflurbendazole	C14 H10 F N3 O	255.08079	255.08577	0.00498	256.09305
10	9.48	2-Isopropylmalic acid	C7 H12 O5	176.06847	116.04652	-60.02195	175.06038
11	9.90	(4-benzoylpiperidino)[4-(tert-butyl)phenyl]methanone	C23 H27 N O2	349.20418	387.1537	37.94952	388.16098
12	11.36	L-Tyrosine	C9 H11 N O3	181.07389	181.07398	0.00008	182.08125
13	11.89	(2Z)-6-hydroxy-2-[(4-hydroxy-3-methoxyphenyl)methylidene]-2,3-dihydro-1-benzofuran-3-one	C16 H12 O5	284.06847	284.06777	-0.0007	285.07489
14	12.96	Isophthalic acid	C8 H6 O4	166.02661	184.03614	18.00953	165.01828
15	13.53	Formononetin	C16 H12 O4	268.07356	268.07335	-0.00021	267.06607
16	14.51	4-Hydroxymandelic acid	C8 H8 O4	168.04226	168.04468	0.00242	167.0374
17	14.87	Palmitic Acid	C16 H32 O2	256.24023	273.26628	17.02605	274.27356
18	16.44	(15Z)-9,12,13-Trihydroxy-15-octadecenoic acid	C18 H34 O5	330.24062	330.2404	-0.00022	329.23312
19	17.38	Prostaglandin F1 α	C20 H36 O5	356.25627	356.25621	-0.00007	355.24893
20	18.17	Palmitoleic Acid	C16 H30 O2	254.22458	276.20885	21.98427	277.21613
21	19.39	9-Oxo-10(E),12(E)-octadecadienoic acid	C18 H30 O3	294.21949	294.21943	-0.00006	295.22668
22	20.11	Androsterone	C19 H30 O2	290.22458	272.21364	-18.01094	255.2103
23	20.49	6-Deoxy- α -L-mannopyranosyl-(1 $\rightarrow$ 3)-[β -D-glucopyranosyl-(1 $\rightarrow$ 6)]-1-O-[3,2'-dihydroxy-27,28-dioxolup-20 (29)-en-28-yl]- β -D-glucopyranose	C48 H76 O19	956.49808	956.49867	0.00059	955.49139
24	21.43	1-Linoleoyl glycerol	C21 H38 O4	354.27701	354.27663	-0.00038	377.26584
25	22.14	(-)-Spiculiporic acid	C17 H28 O6	328.18859	328.18864	0.00005	327.18146
26	23.54	Tridemorph	C19 H39 N O	297.30316	297.30328	0.00011	298.31055
27	24.74	Stearamide	C18 H37 N O	283.28751	283.28758	0.00007	284.29486
28	26.42	Oleanolic acid	C30 H48 O3	456.36035	456.36026	-0.00008	455.35327

min, with a volumetric flow rate of 0.30 mL per minute. The data were initially collated using Compound Discoverer 3.3 (Thermo Fisher) and searched for comparisons through the mzCloud database. The LC-MS

profiles of the sample solutions are presented in Fig. 1, and the list of constituents detected is shown in Tables 2 and 3. The full list of constituents is given in the supplementary material.

Table 3
Top 15 results of LC-MS SXD composition identification.

Name	Formula	Annotation MW	Calc. MW	Annot. DeltaMass	RT (min)	mzCloud Best Match
L-Phenylalanine	C9 H11 N O2	165.07898	148.05239	−17.02659	5.299	100
Nicotinic acid	C6 H5 N O2	123.03203	123.03238	0.00035	2.172	100
Citraconic acid	C5 H6 O4	130.02661	130.02531	−0.0013	4.481	100
Adenosine	C10 H13 N5 O4	267.09675	267.09623	−0.00053	4.597	100
2'-Deoxyadenosine	C10 H13 N5 O3	251.10184	251.10178	−0.00006	4.845	100
Methoxyacetyl fentanyl	C22 H28 N2 O2	352.21508	352.22167	0.00659	16.483	99.9
DL-Tryptophan	C11 H12 N2 O2	204.08988	204.08969	−0.00019	7.361	99.9
2,3,4,9-Tetrahydro-1H-β-carboline-3-carboxylic acid	C12 H12 N2 O2	216.08988	216.09011	0.00023	8.647	99.9
Benzoic acid	C7 H6 O2	122.03678	122.03549	−0.00129	9.79	99.9
Formononetin	C16 H12 O4	268.07356	268.07335	−0.00021	13.528	99.9
L-Isoleucine	C6 H13 N O2	131.09463	131.09467	0.00004	2.586	99.9
Ononin	C22 H22 O9	430.12638	430.12575	−0.00063	13.525	99.9
DL-Homoserine	C4 H9 N O3	119.05824	119.05854	0.0003	1.27	99.9
Benzoic acid	C7 H6 O2	122.03678	122.0355	−0.00128	12.358	99.9
Methoxyacetyl fentanyl	C22 H28 N2 O2	352.21508	320.19586	−32.01922	16.943	99.9

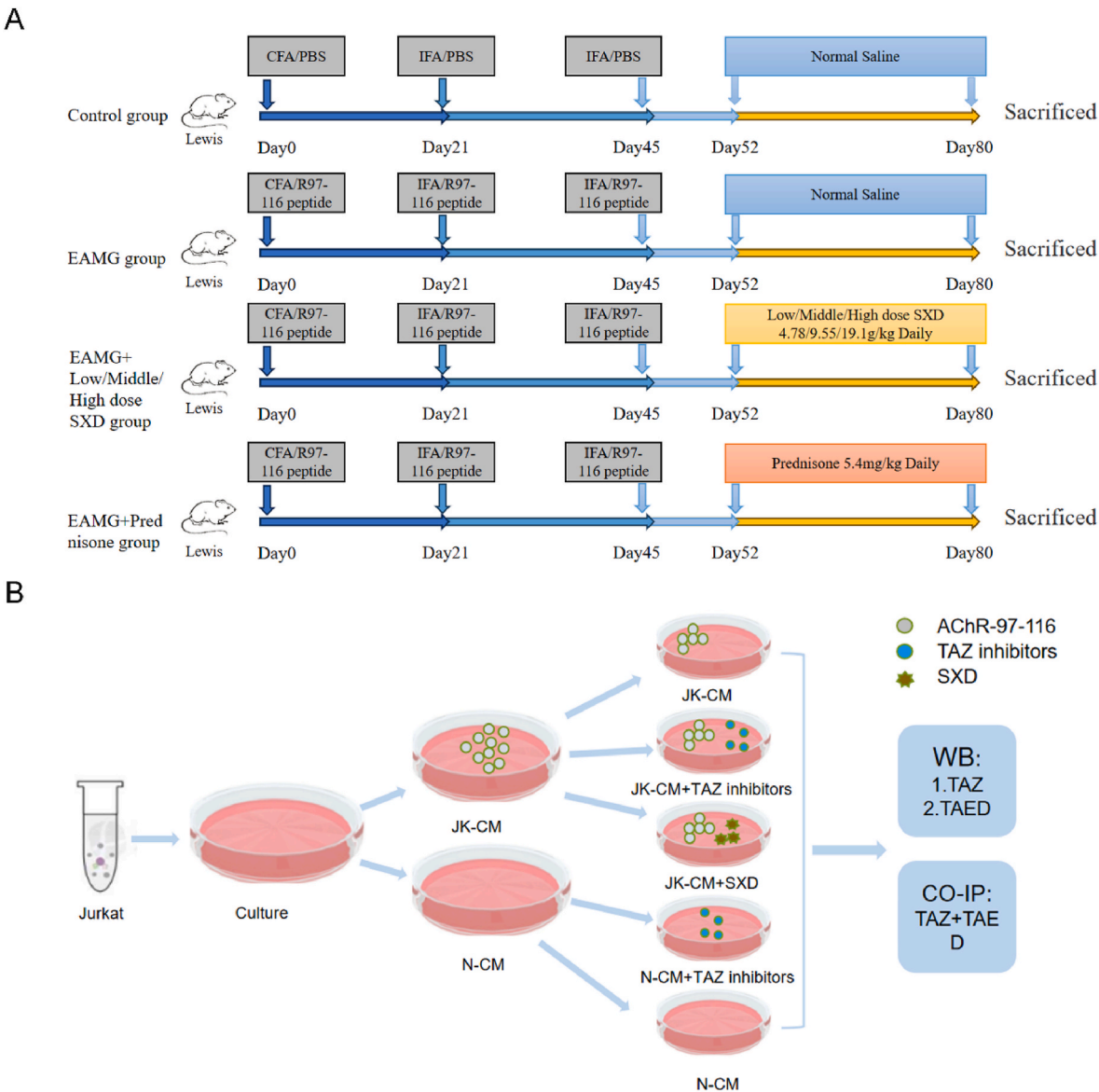


Fig. 2. Experimental technical route.
A. Modelling and experimental design. CFA: Complete Freund's adjuvant; IFA: Incomplete Freund's adjuvant; SXD: Shengxian Decoction; PBS: Phosphate-buffered saline.
B. Culture of Jurkat cells. JK-CM: Jurkat Conditioned Medium; N-CM: Non-AChR-stimulated cell Culture Medium; TAZ: Transcriptional coactivator with PDZ-binding motif.

2.3. Animals and experimental design

Sixty female Lewis rats (4–6 weeks, 150 ± 10 g) obtained from Vital River (Beijing, China) were housed in a specific-pathogen-free environment (temperature 20–26 °C and humidity 40–60 %) with free access to water and food. All animals were randomly caged with five animals per cage and acclimatized for one week before the start of the experiment. Healthy rats were randomly selected as a control group ($n = 10$) and the remaining animals were modeled. The experimental study protocols were reviewed by the Ethics Committee of Nanjing University of Chinese Medicine (Approval number: 202303A025).

Reference the classic method (Baggi et al., 2004; Li et al., 2014; Theissen et al., 2024) to establish a rat model of EAMG. At the time of the first immunization, complete Freund's adjuvant (CFA) (Sigma-Aldrich, USA) containing R97-116 polypeptide (DGDFAIKFKTKVLLDYTGHI) (ChinaPeptides, China), Mycobacterium tuberculosis (strain H37Ra) (BD, USA) and phosphate-buffered saline (PBS) was fully emulsified into an emulsion using a porous corkscrew, and injected subcutaneously at multiple sites in the back, the shoulders, limbs, groin, base of the tail, and plantar surfaces of the feet of the model rats. Each rat was injected with 200 μ L of emulsion (containing 75 μ g of R97-116 peptide and 1 mg of H37Ra bacterial powder), and the control group was injected with the same amount of emulsion prepared by mixing CFA with PBS. Immunization was enhanced by injecting the same dose of emulsion of incomplete Freund's adjuvant (IFA) mixed with R97-116 peptide and PBS at the same site on days 21 and 45, and the adjuvant control group was injected with an equal amount of emulsion of IFA mixed with PBS. One week after completion of the last immunization induction, the model was evaluated based on the behavioral performance of the animals and combined with RNS data and AChR-Ab levels. Grade 2 and higher rats with overt manifestations of myasthenia gravis were considered successful EAMG animal models for modeling. A total of 35 out of 50 rats met the inclusion criteria and the model success rate was 70 %. After model evaluation, EAMG rats were randomly divided into five groups: the model group, the low, medium and high SxD groups, and the positive control prednisone group, with seven rats in each group. After the 52nd day of modelling, different groups were given corresponding drug gavage interventions, the adjuvant control group and the model group were given 0.9 % sodium chloride solution, the low, medium and high dose SxD groups were given different doses of SxD interventions (the concentration of the raw drug was 0.5/1/2 g/mL, 1.76/3.51/7.02 g/kg Daily), and the positive control group received prednisone intervention (0.54 mg/mL, 5.4 mg/kg daily). The rats were sacrificed at day80 and the material collected for subsequent experiments. Weight changes of the rats were recorded weekly during the modelling period and the detail of grouping and experimental design are shown in Figure (Fig. 2A).

2.4. EAMG behavioral grading

The model animals were graded according to the Lennon's method at one week after the third immunization (Lennon et al., 1985). The behavioral performance of the modeled animals was observed at all times during the modeling process, and the model was evaluated once a week to record the data and compare the group to the control group. The evaluation criteria were as follows: Grade 0: showed no obvious signs of muscle weakness; Grade 1: rats with reduced limb strength, easily fatigued, and with reduced activity compared to the control animals; Grade 2: weak bite force, slipping when walking on smooth flat ground, with markedly reduced activity, and difficulty of grasping objects; Grade 3: Slender body size, no obvious resistance to grasping, obvious weakness in tearing, difficulty eating, difficulty supporting the body on table tops, with a high hump on the back, exposed hind limbs, and body shaking when walking; and Grade 4 was given to those whose symptoms were particularly severe during the modelling period and resulted in death. Rats with grade 2 and above, excluding individual cases of death,

were included in the experiment as successful EAMG model ($n = 35$).

2.5. Repeated nerve stimulation (RNS)

Rats were anesthetized and restrained with isoflurane through a liquid gas anesthetic machine, four acupuncture needles were used as electrodes, electrode 1 was inserted into the muscles around the sciatic nerve of rats as a stimulation electrode, electrode 2 was pierced subcutaneously into the abdomen of the rats as a reference electrode, and electrodes 3 and 4 were pierced into the gastrocnemius muscle and Achilles tendon attachments on the ipsilateral side of electrode 2 used as recording electrodes respectively (Supplementary Figure 1A). After connecting the power, and then electrical stimulation was applied at a frequency of 5 Hz for 5 consecutive times and repeated 3 times. The attenuation value D (%) of the action potential was calculated for each time, $D = (1st \text{ action potential} - 5th \text{ action potential}) / 1st \text{ action potential} \times 100 \%$, and $D > 10 \%$ was recorded as positive.

2.6. Enzyme-linked immunosorbent assay (ELISA)

Blood was collected from the tail vein at two time points: one week after the third immunization and one week after the end of the gavage, and whole blood was allowed to stand for 1 h at room temperature. Then the blood was centrifuged at 3000 rpm for 15 min at 4 °C, and the upper layer of serum was retained for subsequent experiments. The expression of AChR-Ab, IFN- γ , and IL-4-related inflammatory factors in serum was detected according to the instructions of the ELISA kit (Aifang Biotech, China).

2.7. Flow cytometry (FCM)

At the end of the pharmacological intervention, fresh thymus tissue was harvested from the animals in each group to prepare single cell suspension. The cell suspensions were co-cultured with leukocyte stimulants (BD Bioscience) at 37 °C, 5 % CO₂ for 4 h. The activated cell suspensions were treated with membrane disruptors and osmotic buffers (all from BD Bioscience) for cell fixation and permeabilization. The treated suspension was stained and labeled at 4 °C with various combinations of fluorescent antibodies for intracellular factors, including antibodies such as anti-CD4-BB700, anti-IFN- γ -FITC, anti-IL-4-PE, anti-IL-17A-APC (all from BD Bioscience); anti-CD4-FITC, anti-CD25-APC, anti-Foxp3-PE-cy7 (all from Thermo Fisher). The prepared cell suspensions were detected using a Moflo XDP ultrafast flow cytometer (Beckman, USA). Gating selection of T cell subpopulations was performed using Flowjo software (BD Bioscience, USA), and the frequency of each cell subpopulation was calculated.

2.8. Hematoxylin-eosin staining (HE)

Fresh thymic tissues from rats were immersed in 4 % paraformaldehyde solution and fixed for 48 h. The samples were dehydrated with different concentrations of ethanol from low to high, and the dehydrated samples were infiltrated with xylene and then embedded by paraffin embedding machine (SAKURA, Japan). The embedded samples were cut into 4–6 μ m sections using a microtome for subsequent experiments.

Sample sections were processed in a constant temperature water bath at 37 °C, dried, deparaffinized in xylene and hydrated with ethanol solutions of different solubilities. The treated sections were stained with hematoxylin-eosin solution (0.5 %–1 %), dehydrated by anhydrous ethanol and permeabilized by xylene, then sealed in neutral resin and placed under a high magnification microscope.

2.9. Immunohistochemistry (IHC)

Sample sections were deparaffinized and hydrated and placed in

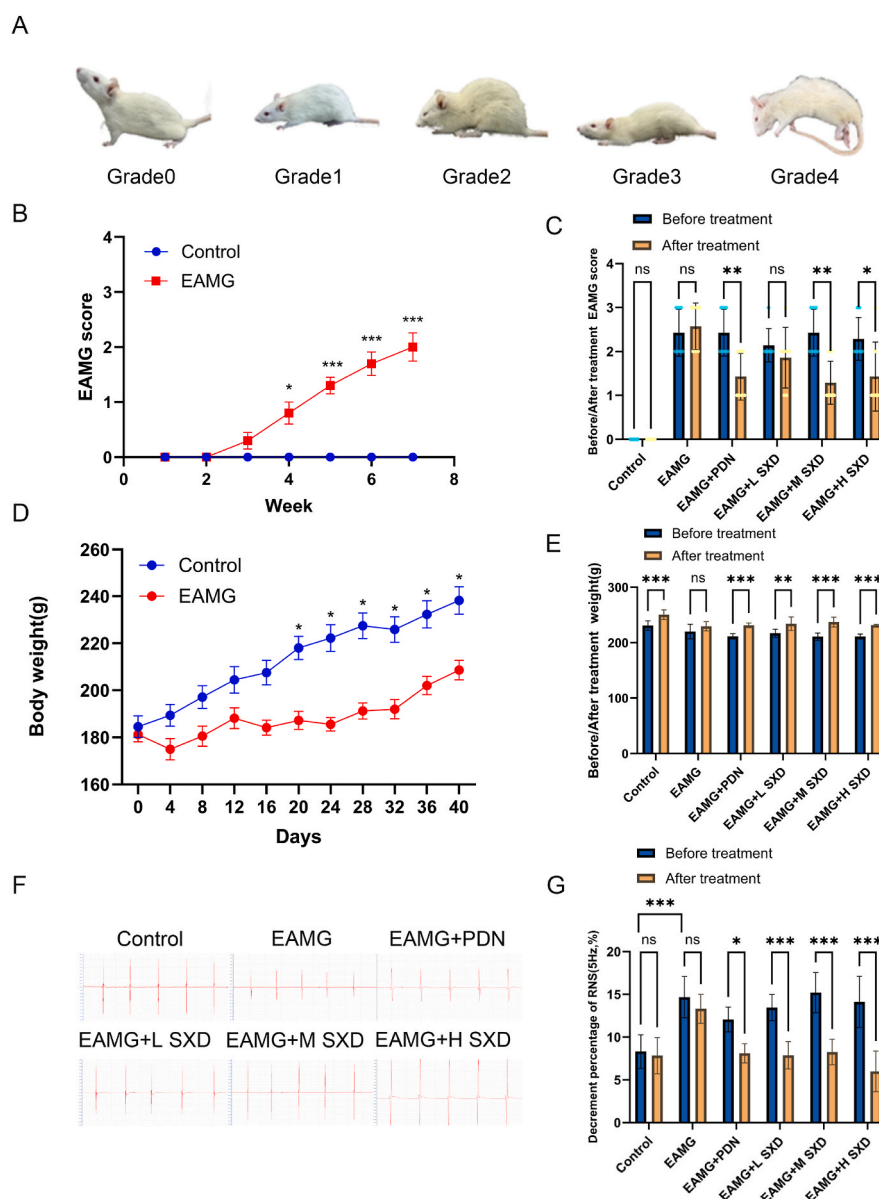


Fig. 3. SxD improves behavioural performance in EAMG rats.

A. Epiograms of model rats at different levels.

B. Changes in EAMG scores during modelling. (n = 10)

C. Comparison of EAMG scores before and after treatment. (n = 7)

D. Rat body weight changes during modelling. (n = 10)

E. Body weight changes in rats before and after treatment. (n = 7)

F. Representative RNS images for each group.

G. Changes in RNS decay rate in each group before and after pharmacological intervention. (n = 7)

The experimental data were expressed as mean $\pm$ SEM, in which the experimental data in Figures B and D were analyzed by unpaired *t*-test, denoted as **p* < 0.05, ***p* < 0.01, ****p* < 0.001, n = 10/group; whereas in Figures C, E and G, the rates of RNS decay in the rats of the control group and the model group before drug intervention were compared by unpaired *t*-test, and the changes in each group before and after intervention were observed by paired *t*-test, and the results were noted as **p* < 0.05, ***p* < 0.01, ****p* < 0.001, n = 7/group.

antigen repair solution containing EDTA for antigen repair. Then 3 % BSA was used to seal the sections at a constant temperature (37 °C, 30 min). After sealing, the target antibodies TAZ (1:400, Thermo Fisher), TEAD (1:200, Thermo Fisher), and FOXP3 (1:400, Thermo Fisher) were incubated separately with the sections in a incubated overnight (4 °C). After washing the sections, secondary antibodies of the corresponding species were added and co-incubated (37 °C, 40 min). Sections were stained with DAB staining solution and nuclei were restained with hematoxylin, then were placed under a microscope and photographed, and the expression of each protein group was calculated using ImageJ

software.

2.10. Immunofluorescence (IF)

Sections were processed as previously described. Sections were fixed for 1 h at room temperature and then co-incubated with rabbit anti-TAZ (1:1000, Proteintech) and rabbit anti-TEAD (1:1000, Proteintech) overnight at 4 °C. Incubation with anti-rabbit fluorescent secondary antibody IgG (1:500, Proteintech) was performed for 1 h at room temperature, followed by DAPI staining for 5 min. Sections were washed 3

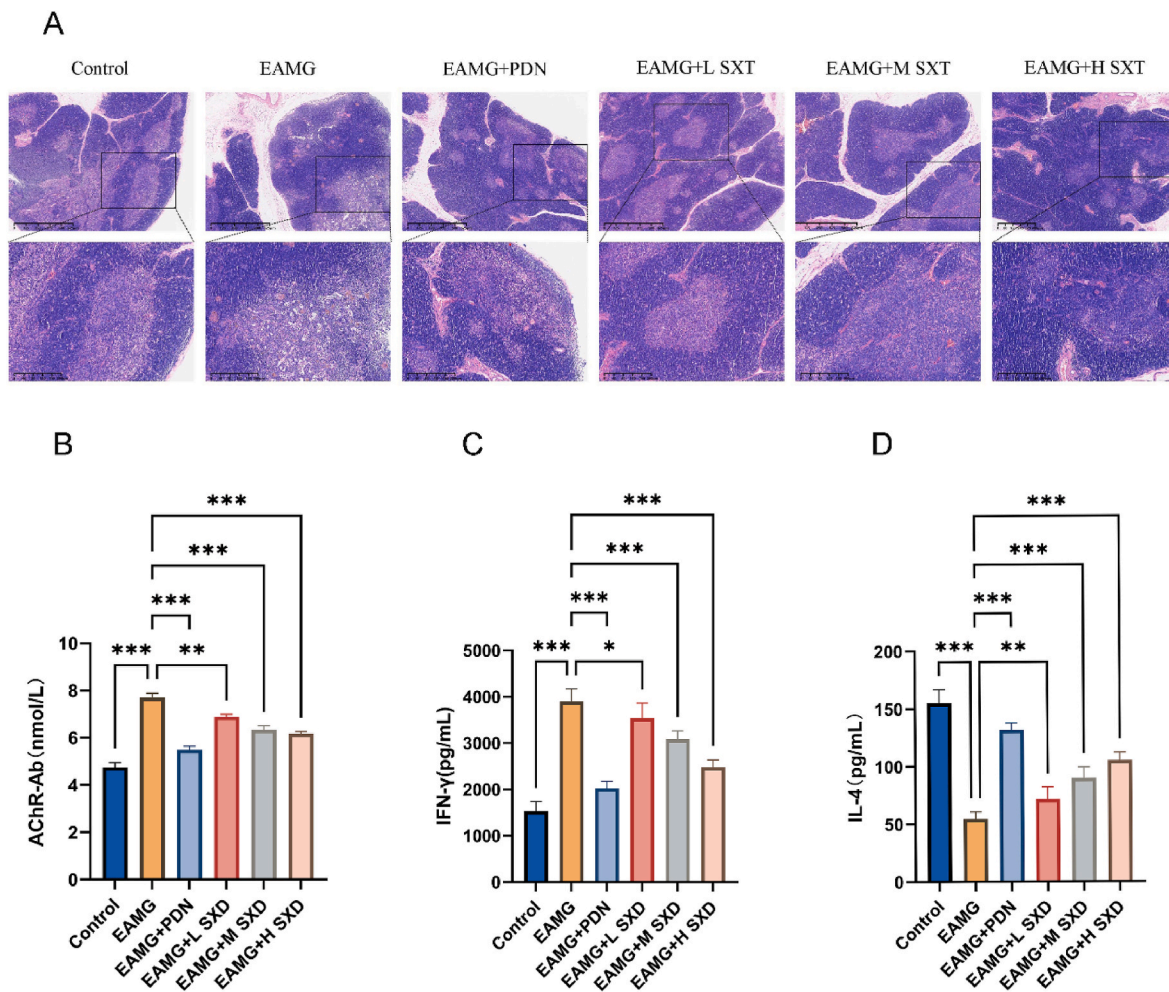


Fig. 4. SxD reduces inflammation in EAMG rats.

A. HE staining results of thymic tissues from different groups of rats under 10x microscope.

B. Determination of anti-AChR expression levels in blood.

C. Determination of IFN-γ expression levels in blood.

D. Determination of IL-4 expression levels in blood.

Data are expressed as means ± SEM. One-way ANOVA test was performed between the six groups, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, $n = 6$ /group.

times with PBS between each of the above steps. Sections were scanned using a fluorescence microscope and fluorescence grey values were quantified using ImageJ software.

2.11. Western Blot (WB)

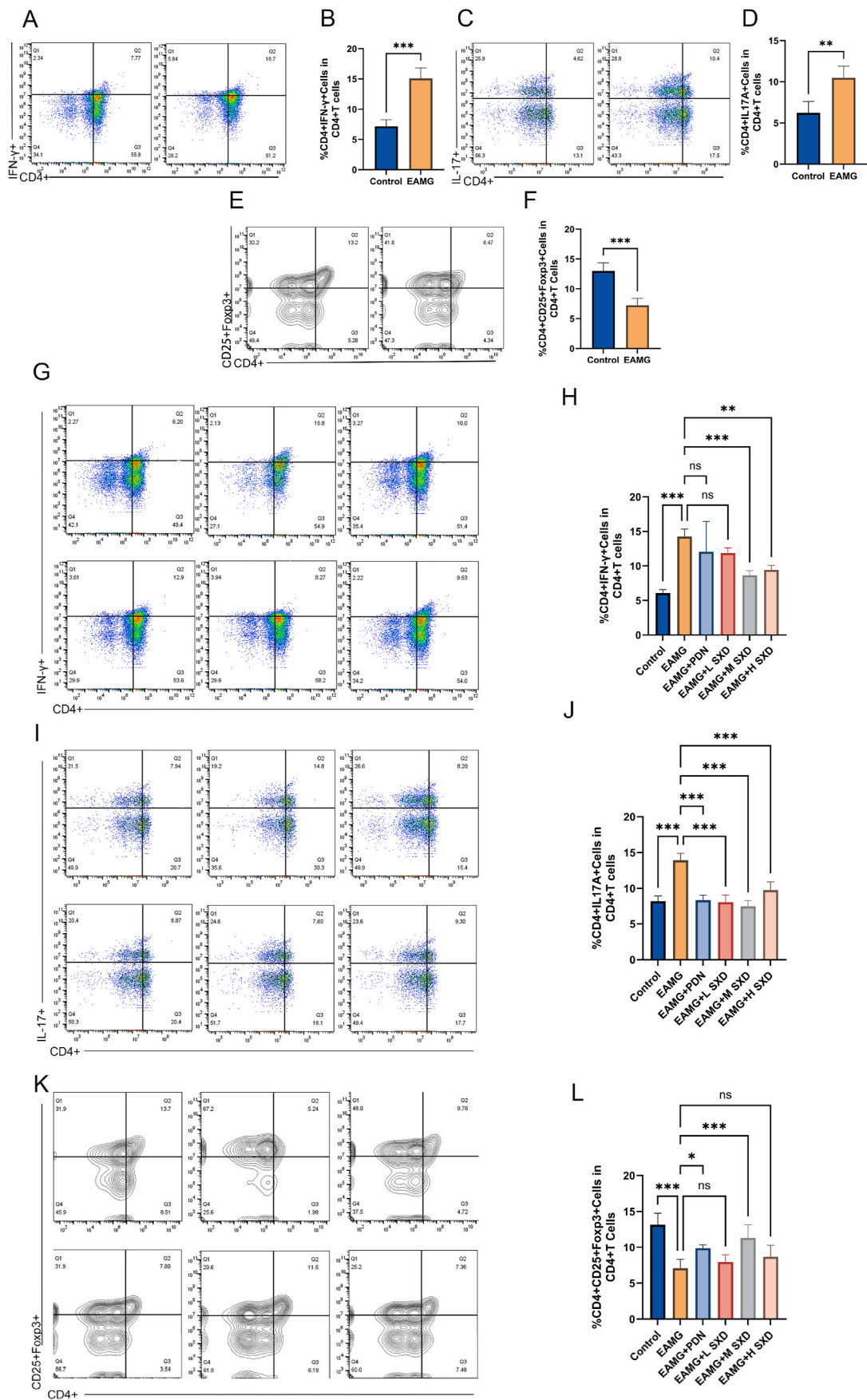
Total protein was extracted from rat Lewis thymus and Jurkat cells using RIPA buffer containing phosphatase and protease inhibitors under ice-cold conditions. Total protein concentration was determined using Nanodrop software. Proteins were separated by 8–10 % SDS-PAGE electrophoresis, followed by transfer to a PVDF membrane and incubation with a protein-free blocking solution for 30 min at room temperature. Primary antibodies were incubated overnight at 4 °C and included TAZ (1:1000, Thermo Fisher), TEAD (1:2000, Thermo Fisher), FOXP3 (1:1000, Thermo Fisher), MST1/2 (1:2000, Affinity), LATS1/2 (1:1000, Affinity) and GAPDH (1:10,000, Proteintech). After washing with 1xTBST (10 min, 3 times), the membranes were co-incubated with secondary antibodies for 1 h at room temperature. The bands were observed and photographed using a chemiluminescence detection system, and the grey scale values of the bands were quantified using ImageJ software.

2.12. Co-immunoprecipitation (Co-IP)

Total proteins from rat Lewis thymus and Jurkat cells were extracted under ice-cold conditions using RIPA buffer containing phosphatase inhibitor and protease inhibitor, and a fraction of the proteins was retained as a blank control. Immunoprecipitation was performed using a Co-immunoprecipitation kit (ACE, China) and pre-treated Protein A + G magnetic beads with primary antibody were added and co-incubated at 4 °C for 2 h. Bound proteins were washed with an elution solution, completely mixed with the loading buffer (BD, USA) and analyzed by WB assay.

2.13. Culture and stimulation of Jurkat cells

Human T-lymphoblastic leukemia Jurkat cells (Clone E6-1) were purchased from Zhejiang Baidi Biotechnology Co. Jurkat (Clone E6-1) was cultured in DMEM (Baidi, China) medium supplemented with 10 % FBS (Gibco, USA) and 1 % penicillin/streptomycin (Gibco, USA) in an incubator containing 5 % CO₂ at 37 °C. Jurkat (clone E6-1) was co-incubated with AChR for 24–48 h to obtain Jurkat Conditioned Medium (JK-CM), and Non-AChR-stimulated cell Culture Medium (N-CM) was used as a negative control. Jurkat was starved overnight in serum-free medium, and subsequent experimental manipulations were



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Fig. 5. SXD showed immunosuppressive effects similar to PDN in the EAMG rat model.

- A. Percentage of Th1 cells labeled with CD4+IFN- γ + by flow cytometry.
- B. Th1 cell expression in the model and control groups.
- C. Percentage of Th17 cells labeled with CD4+IL17+ by flow cytometry.
- D. Th17 cell expression in the model and control groups.
- E. Percentage of Treg cells labeled with CD4⁺CD25+FOXP3+ by flow cytometry.
- F. Treg cell expression in the model and control groups.
- G. Flowchart of Th1 expression in each group after drug intervention.
- H. Th1 expression in each group after pharmacological intervention.
- I. Flowchart of Th17 expression in each group after drug intervention.
- J. Th17 expression in each group after pharmacological intervention.
- K. Flowchart of Treg expression in each group after drug intervention.
- L. Treg expression in each group after pharmacological intervention.

The experimental data were expressed as mean $\pm$ SEM, in which the experimental data in Figures B, D, and F were tested by unpaired *t*-test, denoted as **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *n* = 5/group; while the experimental data in Figures H, J, and L were analyzed in the respective groups by one-way ANOVA, and the results were denoted as **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *n* = 5/group.

performed after a 12h intervention with TAZ inhibitors (Fig. 2B).

2.14. Safety assessment of the SXD

After 4 weeks of gavage intervention, we collected serum from animals in each group (blank, model, SXD, and PDN groups) with five samples from each group. Alanine aminotransferase (ALT) and aspartate aminotransferase (AST) were used as indicators of liver function, and creatinine (CREA) and blood urea nitrogen (BUN) were used as indicators of renal function to evaluate the safety of long-term administration of SXD and PDN.

2.15. Statistical analyses

The experimental data were statistically analyzed using Graphpad prism 9.0 software and all data are expressed as mean $\pm$ SEM. Two different samples were analyzed with Student's *t*-test, two-tailed. For comparisons of more than two groups, one-way ANOVA followed by Tukey's post hoc test was applied. Moreover, a two-way ANOVA was employed to make a comparison between two sets of continuous data. A *p* < 0.05 between data was recognized as a significant difference, denoted as **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

3. Results

3.1. SXD improved behavioral performance in EAMG rats

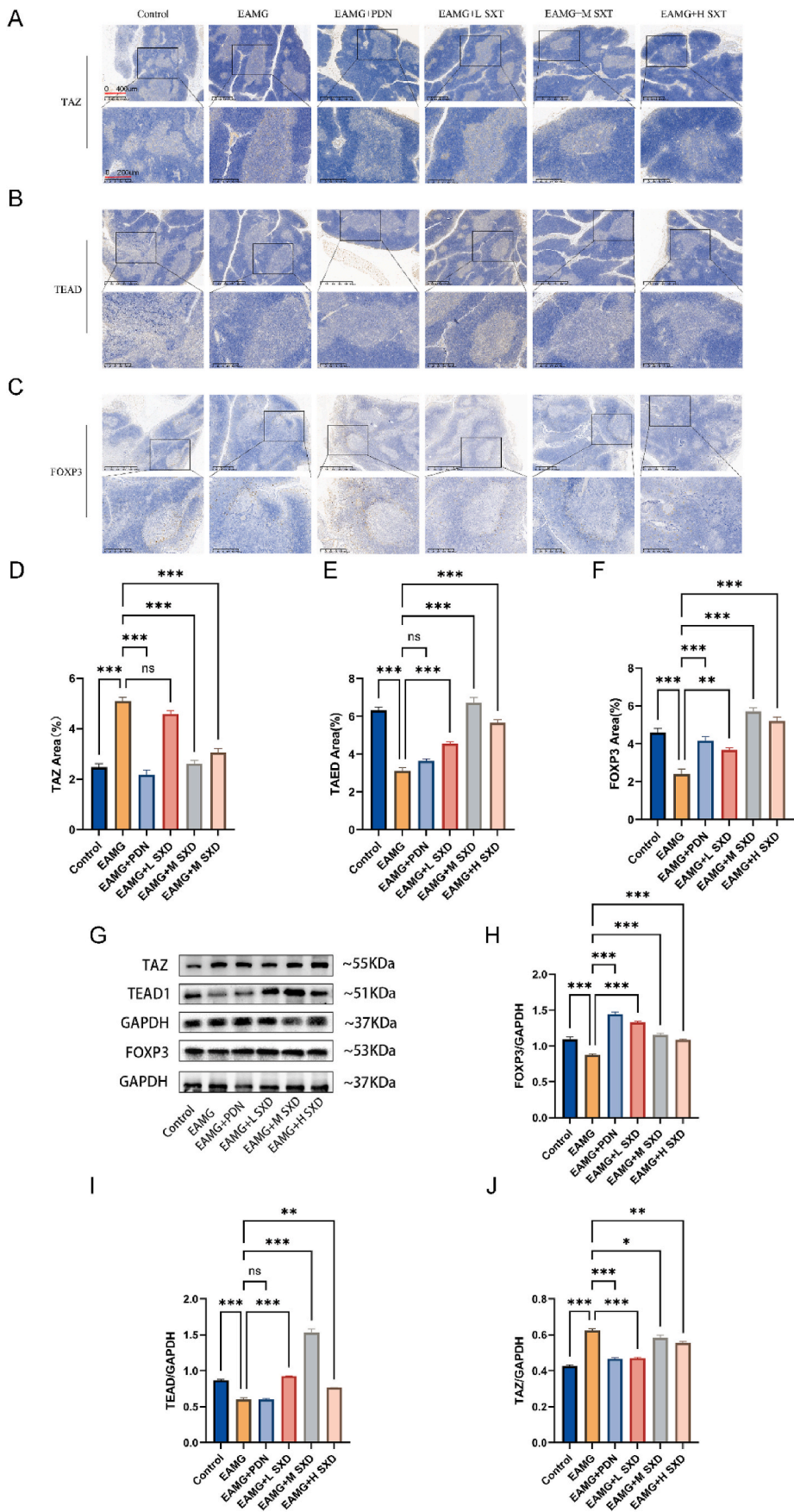
In order to examine the therapeutic effect of SXD on MG, we used the Lennon grading method to score the model animals based on their behavior and clinical signs at the end of modeling and treatment. The representative images of animals at each level are shown below (Fig. 3A). By week 3, the modeled rats began to show signs of MG, and by week 4, they showed significant differences from the control animals (Fig. 3B). At the end of treatment, the clinical performance of animals in all dose groups improved and the clinical symptom scores decreased significantly (Fig. 3C). During the modelling period, we observed a significant slowdown in the weight gain trend of model rats compared to the control group (Fig. 3D). This phenomenon was improved in the EAMG rats after intervention with different doses of SXD (Fig. 3E). To assess the effect of SXD on the muscle tone of rats, we performed the RNS test on the leg muscles of rats. The results showed that the muscle force decay rate of the model group was higher than that of all other groups, and after the intervention, the muscle force decay rate of the animals in all other groups decreased significantly (Fig. 3F–G; Supplementary Figure 1B). This suggested that SXD may increase muscle strength in model rats and have a therapeutic effect on myasthenia gravis.

3.2. SXD repaired thymus tissue damage and reduced inflammatory response

The thymus is a vital immune organ in rodents and humans. In patients with MG, the thymic tissue is in a state of chronic inflammation for a long time. To observe the occurrence of inflammation *in vivo*, we performed HE staining on the thymus tissue of rats in each group. In the control group, the thymus tissue was structurally intact, morphologically clear, with clear boundaries between the cortex and medulla, and the representative structure, thymic vesicles were clearly visible. In the model group, compared with the control group, the morphology of the thymic tissue was altered, the boundary between the cortex and the medulla was blurred, the medulla was diffusely aggregated and the structure of the thymic vesicles was damaged, suggesting that the tissues had more pronounced inflammatory infiltration. The inflammatory infiltration in the groups was improved after the intervention, with the medium-dose SXD group being the closest to the control group (Fig. 4A). In addition, AChR-Ab, IFN- γ and IL-4 were detected in the peripheral serum of rats in each group by ELISA. Compared with the model group, AChR-Ab and IFN- γ expression decreased and IL-4 expression increased in the SXD and prednisone groups at all doses, and the differences among the SXD dose groups were not significant (Fig. 4B–D). This indicates that both SXD and PDN are able to regulate the inflammation level of EAMG model rats and repair thymic tissue damage to a certain extent.

3.3. SXD restored Treg/Th17 imbalance in thymus induced by EAMG model

Treg/Th17 imbalance in the thymus plays a critical role in the progress of EAMG. As expected, Th1 and Th17 cell subpopulations in the thymus of model group animals were significantly increased after AChR peptide-induced immunization compared to the control group (Fig. 5A–D), which suggests the occurrence of inflammation and autoimmunity in the EAMG model group rats. In contrast, the frequency of Regulatory T cells (Treg), which have the function of regulating immunity and suppressing autoimmunity, decreased significantly in the model group (Fig. 5E–F). This means that the proportion of pro-inflammatory cells is increased and the proportion of anti-inflammatory cells is decreased in the T cells of the model group. In addition, while SXD and PDN significantly reduced the frequency of the Th17 cell subpopulation in the EAMG model *in vivo*, they were able to enhance the expression of the Treg cell subpopulation, and the effect of SXD was superior to that of PDN (Fig. 5G–L). This suggested that the immunotherapeutic effect of SXD is not dependent on a single T-cell subpopulation but focuses on maintaining the balance between Th17 and Treg cell subpopulations. All doses of SXD groups showed immunomodulatory effects, but their intergroup comparisons showed that the effects were more pronounced in the middle and high dose SXD groups.



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Fig. 6. SXD restores EAMG-inhibited signalling in the TAZ/TEAD pathway

A. Expression of TAZ in thymic tissues of rats of all groups under 10x and 20x magnification.

B. Expression of TEAD in thymic tissues of rats of all groups under 10x and 20x magnification.

C. Expression of FOXP3 in thymic tissues of rats of all groups under 10x and 20x magnification.

D. Quantitative analysis of TAZ expression in IHC. (n = 3)

E. Quantitative analysis of TEAD expression in IHC. (n = 3)

F. Quantitative analysis of FOXP3 expression in IHC. (n = 3)

G. Representative Western blot images.

H. Quantitative analysis of FOXP3 expression in WB. (n = 5)

I. Quantitative analysis of TEAD expression in WB. (n = 5)

J. Quantitative analysis of TAZ expression in WB. (n = 5)

One-way ANOVA test was performed between the groups. Data are expressed as means $\pm$ SEM (n = 3 or 5, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

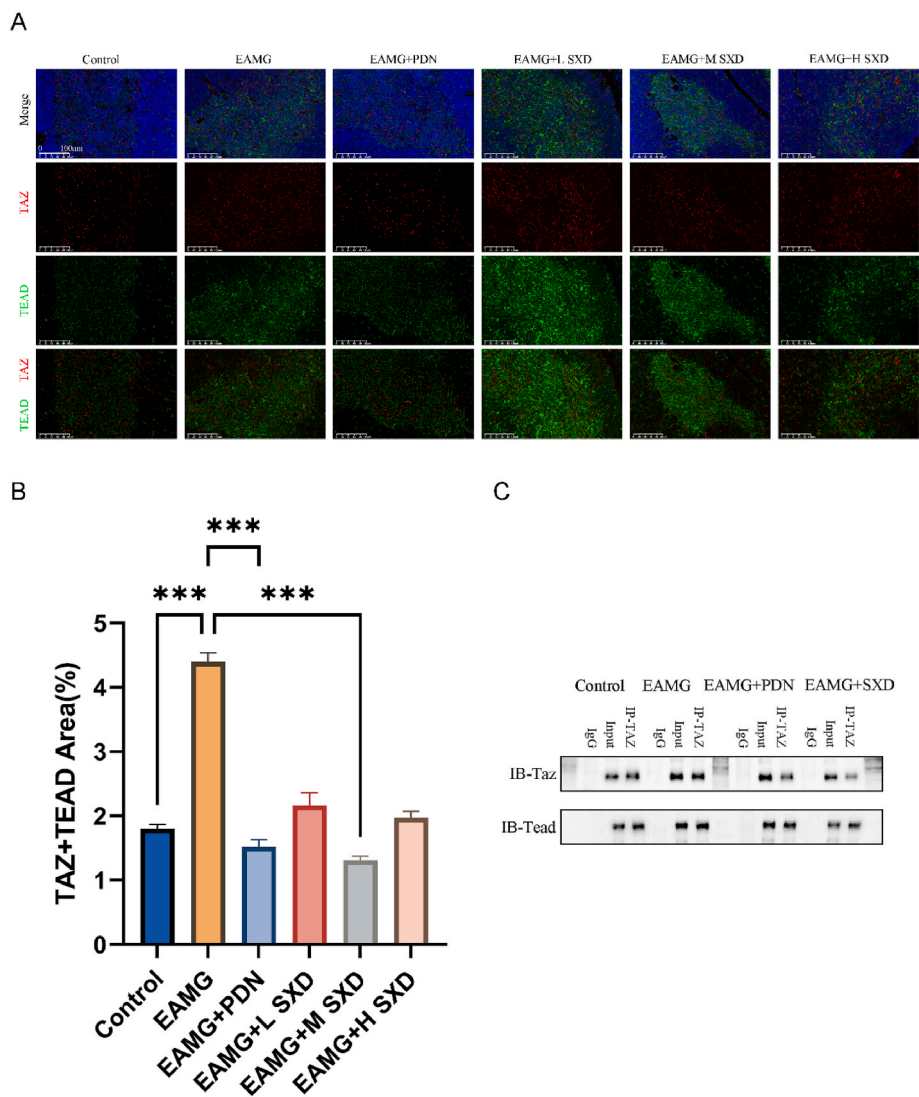


Fig. 7. SXD prevents transcription by reducing the binding of TAZ to TEAD.

A. IF images of TAZ and TEAD co-localization in the thymus at 20x magnification.

B. Quantitative analysis of TAZ binding to TEAD.

C. Representative Western blot image of Co-IP.

One-way ANOVA test was performed between the groups. Data are expressed as means $\pm$ SEM (n = 5, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

3.4. SXD modulated the TAZ/TEAD pathway to attenuate EAMG

The abnormal changes in Th17 and Treg in the model group attracted our attention. It has been shown that the transcriptional co-activator TAZ can have an effect on the proliferation and differentiation of Th17 cells (Geng et al., 2017). Therefore, we took TAZ as core module mediated and the focus of our study. The expression of TAZ and its

downstream proteins TEAD and FOXP3 in rat thymic tissue was investigated by IHC. We also used WB experiments to observe the changes in the proteins after the intervention in each group. We found that TAZ, TEAD and FOXP3 were expressed in the thymic tissues of all groups, with TAZ and TEAD were mostly dispersed in the medulla of the thymic tissues, while FOXP3 was aggregated in the cortical-medullary junction (Fig. 6A–C). Further quantification using ImageJ software revealed that

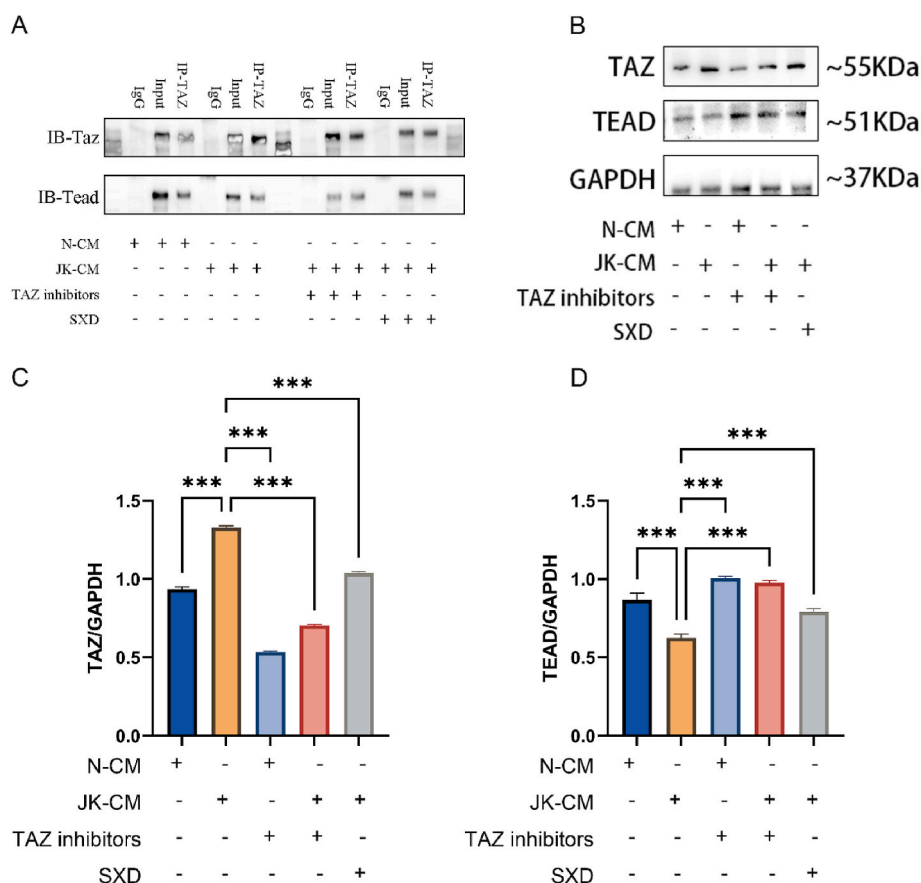


Fig. 8. In vitro validation of the mechanism of SXD for the treatment of EAMG.

A. Representative Western blot image of Co-IP.

B. Representative Western blot images in Jurkat.

C. Quantitative analysis of TAZ expression in Jurkat.

D. Quantitative analysis of TEAD expression in Jurkat.

One-way ANOVA test was performed between the groups. Data are expressed as means $\pm$ SEM ($n = 3$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

the expression of TAZ was significantly downregulated in each intervention group compared to the model group, while the expression of FOXP3 and TEAD was significantly upregulated in the mid-dose SXD group, and it was also upregulated in all other groups (Fig. 6D–F). Observation of the expression of various proteins in the TAZ/TEAD pathway by WB experiments also showed that SXD increased the expression of FOXP3 and TEAD and decreased the expression of TAZ in the thymus tissue of EAMG rats (Fig. 6G–J). This implied that SXD can regulate the expression of relevant proteins in the TAZ/TEAD pathway, which may be a potential mechanism for its therapeutic effects. Combining the results of IHC and WB experiments, the modulation of TAZ/TEAD pathway proteins by medium and high doses of SXD was significant, and the effect of low doses of SXD was not stable. The results of the comparison between the mid-dose and high-dose SXD groups showed better results in the mid-dose group.

3.5. SXD inhibited the binding of TAZ to TEAD

Based on the detection of significant differences in TAZ/TEAD pathway-related proteins between the groups, we used IF experiments to explore the corresponding mechanisms and verified them by Co-IP experiments. Both TAZ and TEAD were found to be expressed in the thymic medulla with overlapping localization, suggesting that TAZ and TEAD may bind (Fig. 7A). Quantitative analysis of the binding fraction revealed that the binding of TAZ to TEAD was significantly increased in the model group compared to the control group, and the mid-dose SXD group was able to reduce the binding between the two after intervention

(Fig. 7B). The experimental results obtained after quantification by precipitation of TAZ with TEAD in thymic tissue were in accordance with the IF experiments (Fig. 7C). This means that SXD was able to prevent the binding of TAZ to TEAD, which may be its potential mechanism of action.

3.6. Inhibiting the co-transcription factor TAZ is a central mechanism in treating EAMG with SXD

To investigate whether the co-transcription factor TAZ is a central mechanism of SXD in regulating T cells to treat MG, we infused TAZ inhibitors into a Jurkat cell line to reduce TAZ expression in the cells. Jurkat cells are suspension cells derived from T lymphocytes from patients with acute T-cell leukemia and are commonly used in studies of T-cell signalling. The effect in vitro was assessed by intervening cells with drug-containing serum from the medium dose SXD group. WB and Co-IP assays were used to monitor TAZ and TEAD expression and binding. Binding of TAZ to TEAD was detected in all groups, although there was a decreasing trend in the binding of both after intervention, there was no significant difference (Fig. 8A). The reason for this phenomenon may be that dynamic post-translational modifications present in the in vivo setting (e.g., phosphorylation/ubiquitination modifications of TAZ) may not be fully reproduced in the simplified in vitro system, and the in vitro experimental system lacks the complex microenvironmental regulatory network in vivo (e.g., paracrine signaling, extracellular matrix interactions, etc.), which may also lead to different features of TAZ-TEAD binding kinetics. TAZ expression was found to be significantly

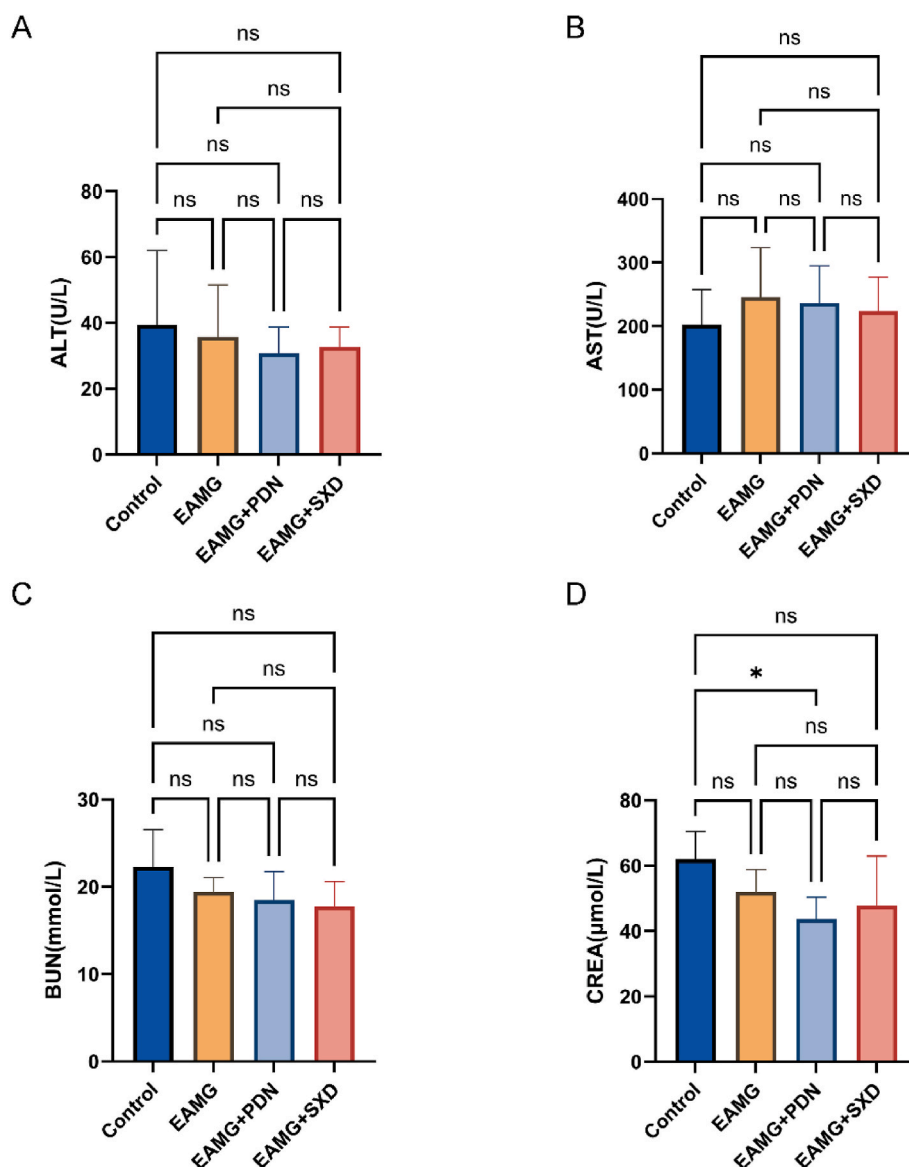


Fig. 9. Observations on liver and kidney functions of rats in various groups after 4 weeks of gavage.

A. Levels of alanine aminotransferase in rats in various groups after gavage (Reference Point: 21.53–61.75).

B. Levels of aspartate aminotransferase in rats in various groups after gavage (Reference Point: 41.47–250.65).

C. Levels of creatinine in rats in various groups after gavage (Reference Point: 9.75–22.71).

D. Levels of urea nitrogen in rats in various groups after gavage (Reference Point: 10.90–118.07).

One-way ANOVA test was performed between the groups. Data are expressed as means $\pm$ SEM ($n = 5$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

downregulated in the single inhibitor group compared to the control group, suggesting that TAZ inhibitors could reduce TAZ expression in Jurkat cells (Fig. 8B). Both TAZ inhibitor and SXD intervention significantly reduced TAZ expression and increased TEAD expression compared to the model group (Fig. 8C–D). This indicated that SXD has the same effect as TAZ inhibitors, but its reduction of TAZ binding to TEAD in vivo may be influenced by the organismal environment in vivo.

3.7. Long-term use of SXD has not caused liver or kidney damage

The results showed that the liver and kidney function parameters of all groups were within the reference range. Compared with the blank control group, no significant changes in liver and kidney functions were observed in rats that received SXD for a long period of time. This suggests that long-term administration of SXD is safe. In addition, after 4 weeks of PDN administration, creatinine levels in the rats decreased

significantly compared to those in the blank control group, which may be due to muscle wasting and decrease in muscle strength of the rats, but it was still within the normal range (Fig. 9).

4. Discussions

The aim of this study was to investigate the potential of SXD for the treatment of MG and to elucidate the possible mechanism of its effect. Infusion of AChR peptide in vivo is a commonly used method to induce autoimmunity in animals to establish an EAMG model. Based on the classical method, we constructed a stable EAMG animal model by improving the selection of drug dosage and injection site through preliminary experiments. After 4 weeks of SXD and PDN administration, the body weight of EAMG rats gained slowly and the rate of muscle strength attenuation was significantly reduced, indicating that both SXD and PDN could improve the clinical manifestations of EAMG rats and had

certain therapeutic effects. The staining results of thymic tissues and serum inflammatory factor levels showed that both SXD and PDN were able to reduce the levels of anti-AChR and IFN- γ and repair damage to thymic tissues. This indicates that SXD has the similar anti-inflammatory effect as PDN. In FCM experiments, SXD proved to increase the frequency of Treg cell differentiation while being able to decrease the frequency of Th1 and Th17 cells, which indicates that SXD has an immunosuppressive effect similar to that of PDN. We detected the expression of TAZ, TEAD and FOXP3 proteins in thymic tissues using IHC and found that SXD was able to significantly downregulate the expression of TAZ and upregulate the expression of FOXP3 and TEAD in EAMG thymus, which was consistent with the FCM experiment and was further verified in the WB experiment. In IF and Co-IP experiments, SXD was observed to reduce the binding of TAZ to TEAD. In contrast, the regulation of TAZ/TEAD pathway related proteins by PDN was not significant and consistent. Therefore, the mechanism of action of SXD in the treatment of EAMG through the TAZ/TEAD pathway was basically elucidated. To further verify whether TAZ is the core mechanism by which SXD exerts its therapeutic effect, we inhibited the expression of TAZ in Jurkat cells using TAZ inhibitors and treated with various drugs, and the results also showed that SXD inhibited TAZ expression in Jurkat cells and has similar effects as TAZ inhibitors. All experiments suggest that SXD can improve the behavioral performance of EAMG, by inhibiting inflammation and regulating the body's immunity through the TAZ/TEAD signaling pathway.

The treatment of autoimmune diseases has been dominated by immunosuppression so far (Jung and Kim, 2022; Liebman, 2016). PDN is often used in the treatment of autoimmune diseases considering its anti-inflammatory and immunosuppressive effects (Dasgupta et al., 1991; Liu et al., 2020). However, long-term use of immunosuppressive drugs may lead to a decrease in immunity and cause other complications (Schneider-Gold et al., 2005). Some studies have used isoxazole immunosuppressants to treat EAMG animals, which also improve symptoms in the EAMG animals (Huang et al., 2021; Koseoglu et al., 2023), while they are the same limitations as PDN. Since most patients with MG are accompanied by thymic hyperplasia and thymoma, surgical removal

of the thymus has also become a common clinical treatment for MG. Some studies have claimed that 3 years after thymectomy, the patients tend to show decreased dependence on immunosuppressive drugs and better prognosis (Iorio, 2024). However, it has also been found that patients who undergo thymectomy have higher all-cause mortality and cancer risk than non-thymectomized patients and the healthy population (Kooshesh et al., 2023). TCM treatment of myasthenia gravis has the advantage of being multi-component and multi-targeted (Zhu et al., 2022). It has been demonstrated that SXD contains a variety of pharmacological components that have the capacity to reduce inflammation and modulate immunity. Previous studies have shown that SXD can improve symptoms in rats with idiopathic pulmonary fibrosis and inflammation-induced obstructive lung disease (Liang et al., 2024; Yang et al., 2023). In this study, we found that SXD could improve the behavioral performance of EAMG rats, reduce the expression levels of anti-AChR antibodies and IFN- γ in vivo, and had immunosuppressive effects along with those of PDN. During experimental conduction, we found that intestinal infarction and soft stools were easily observed in animals taking PDN for a long time, while no obvious abnormality was observed in the traditional Chinese medicine group, which indicates that long-term administration of SXD has higher safety than PDN. In addition, Buzhongyiqi Decoction (BZYQD) is also commonly used in the treatment of mild MG (Li et al., 2021; Qiu et al., 2024). Studies have shown that BZYQD improves chronic muscle atrophy by modulating calcium signaling and downregulating NCoR1 protein expression (Zhu, M. et al., 2017). In contrast, the present study showed that SXD improves muscle strength by modulating the TAZ/TEAD pathway to reduce inflammation and restore immune homeostasis. The two mechanisms of action are distinct, and BZYQT has not been reported to regulate immunity and repair tissue damage. In comparison, BZYQD is more applicable to the onset stage of MG, while the immune imbalance caused by late inflammation is the predominant pathological mechanism. Therefore, SXD with stronger anti-inflammatory capacity and immunosuppressive effects is clearly more advantageous. In addition, after 4 weeks of gavage intervention, we collected serum from animals in each group (blank, model, SXD and PDN groups). Alanine aminotransferase

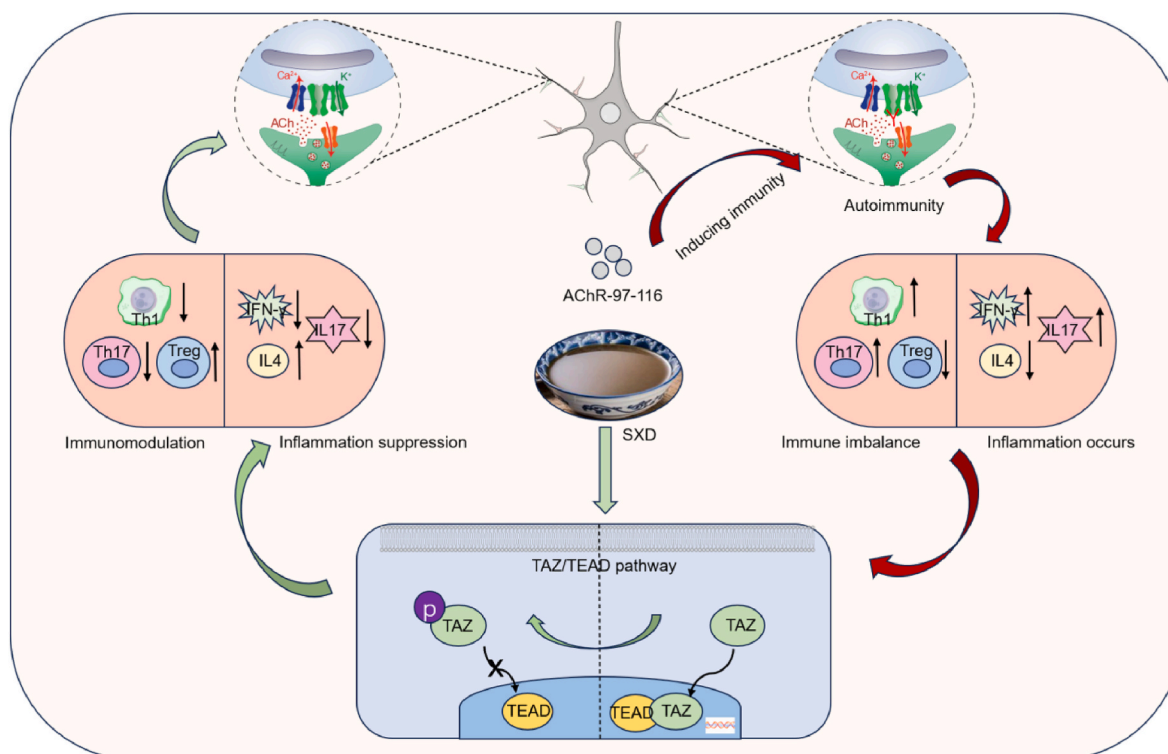


Fig. 10. Mechanism of action of SXD for the treatment of EAMG.

(ALT) and aspartate aminotransferase (AST) were used as liver function indicators, and creatinine (CREA) and urea nitrogen (BUN) were used as kidney function indicators to evaluate the safety of long-term administration of SXD and PDN. The results showed that the liver and kidney function indexes of each group were within the reference range. Compared with the blank control group, no significant changes in liver and kidney functions were observed in rats receiving long-term SXD. It is worth noting that after 4 weeks of PDN administration, the creatinine level of rats decreased significantly compared with that of the blank group, which may be caused by the reduction of muscle and deterioration of muscle strength in rats, but it is still within the normal range. In conclusion, the safety of long-term administration of SXD seems to be higher than that of PDN, but this is only limited to the results of 4 weeks, and the comparison may be more informative if the treatment is continued for a long period of three months or one year.

There are many studies demonstrating changes in T cell subsets in MG (Ashida et al., 2021; Ingelfinger et al., 2021; Peng et al., 2024). This phenomenon was confirmed in our experiments. By FCM experiments, we observed a significant increase in the percentage of Th1 and Th17 cells in EAMG rats, while the expression of Treg, which has an immunomodulatory role, was suppressed, which was reversed by the intervention of SXD. The balance between Th17 and Treg has become the focus of research in autoimmune diseases (Altınönder et al., 2024). The occurrence of abnormal changes in Th17 and Treg balance and the factors influencing their development provide new aspects to elucidate the mechanism of action of SXD in the treatment of MG. It has been reported that the expression of the transcriptional coactivator TAZ is upregulated in initial CD4⁺ T cells during their differentiation into immunogenesis-promoting Th17 cells (Zhu et al., 2023). TAZ activity is regulated by the Hippo signaling pathway, which has been shown to be associated with cell proliferation and differentiation (Kwon et al., 2022; Li and Guan, 2022; Plouffe et al., 2016). Upon disease onset, unphosphorylated TAZ enters the nucleus where it is activated and binds to the co-transcription factor TEAD, which turns on the transcription of downstream genes and promotes the proliferation and differentiation of CD4⁺ T cells into the pro-inflammatory Th17 cell subtype (Hong et al., 2018). In turn, TEAD can promote the proliferation and differentiation of Treg cells (Geng et al., 2017). Some studies have shown that the active ingredients in qi-tonic herbs can regulate the expression of TAZ/TEAD pathway-related proteins (Zhang et al., 2019; Zhu et al., 2021). In our experiments, we found that SXD was able to significantly downregulate the expression of TAZ, upregulate the expression of TEAD and FOXP3, and inhibit the binding of TAZ to TEAD in the thymus tissue of EAMG rats. This suggests that SXD is able to regulate the TAZ/TEAD pathway, which may be a potential mechanism for its treatment of MG. Nevertheless, this study still has some limitations, such as the unclear role of SXD in the upstream and downstream of the TAZ/TEAD pathway and its effect on the acetylation mechanism of FOXP3 are still unclear and will be reported in future studies.

5. Conclusions

In conclusion, SXD can treat EAMG and its mechanism of action may be achieved by activating the TAZ/TEAD pathway, promoting the degradation of the transcriptional co-activator TAZ and inhibiting its entry into the nucleus to bind to TEAD, thereby restoring the balance of the body's immune function. The present study demonstrates the therapeutic effect of SXD on EAMG and its possible mechanism of action, providing support for the clinical treatment of MG (Fig. 10).

CRediT authorship contribution statement

Shida Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. Xiangrui Zhang: Writing – review & editing, Methodology, Investigation, Formal analysis. Yu Bai: Methodology, Investigation, Formal analysis, Data

curation. Jun Shi: Supervision, Methodology. Yan Sun: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. Haoxin Wu: Writing – review & editing, Resources, Project administration, Conceptualization.

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Abbreviations

MG	Myasthenia gravis
SXD	Shengxian Decoction
AChR	Acetylcholine receptor
EAMG	Experimental autoimmune myasthenia gravis
RNS	Repeated nerve stimulation
HE	Hematoxylin-eosin
ELISA	Enzyme-linked immunosorbent
FCM	Flow cytometry
IHC	Immunohistochemistry
IF	Immunofluorescence
TAZ	Transcriptional coactivator with PDZ-binding motif
TEAD	TEA Domain Transcription Factor
FOXP3	Transcription factor Forkhead box P3
WB	Western blotting
Co-IP	Co-immunoprecipitation
Th1	Type 1 helper T cells
Th17	Type 17 helper T cells
Treg	Regulatory T cells
IL-17	Interleukin 17
LC-MS	High-resolution liquid chromatography-mass spectrometry
PBS	Phosphate-buffered saline
CFA	Complete Freund's adjuvant
IFA	Incomplete Freund's adjuvant
PDN	Prednisone
EDTA	Ethylenediaminetetraacetic acid
BSA	Bovine serum albumin
DAB	Diaminobenzidine
IgG	Immunoglobulin G
RIPA	Radioimmunoprecipitation assay buffer
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
CREA	Creatinine
BUN	Urea nitrogen

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.120250>.

Data availability

Data will be made available on request.

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